

An innovative structural energy storage solution using fly ash-cement composites for net-zero energy buildings

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ABSTRACT

The rapid advancement of renewable energy highlights the urgent need for safe, cost-effective, and scalable energy storage solutions, particularly for net-zero energy buildings. In this study, we introduce an innovative energy storage solution utilizing fly ash-cement composites (FCS) as multifunctional components. The FCS, incorporating fly ash as a mineral admixture, achieves a refined pore structure and homogeneous air void distribution during early hydration, leveraging the ball-bearing characteristics and pozzolanic effect of fly ash. The optimized FCS, containing 50 wt% fly ash, exhibited an impressive ionic conductivity of 25.6 mS cm^{-1} and a compressive strength of 5.5 MPa after just one day of curing. When integrated into structural energy storage systems, it delivers a high specific capacitance of 102.4 mAh g^{-1} at 0.1 A g^{-1} , an energy density of 73 Wh kg^{-1} , and a power density of 76.3 W kg^{-1} , maintaining 92.9 % capacitance retention over 2000 cycles. These results underscore the scalability, cost efficiency, and structural benefits of FCS, offering a promising pathway to integrate energy storage directly into buildings and infrastructure. Moreover, this strategy provides a sustainable, high-value application for large volumes of industrial solid waste while addressing energy challenges in extreme environments such as deep-sea, deep-earth, and extraterrestrial constructions.

1. Introduction

The continuous increase in carbon emissions is a primary driver of global climate change, posing severe threats to ecosystems and human health [1]. The construction industry, as a significant contributor, accounts for approximately 40 % of global emissions, with cement production - being an essential building material - responsible for around 8 % of global carbon emissions [2,3]. Consequently, it is urgent to promote eco-friendly, resource-efficient buildings to advance sustainable development and develop green economy. Integrating renewable energy sources to facilitate the development of net-zero energy buildings (NZEBS) has gained widespread recognition as a pivotal objective for future architectural advancements [4,5]. Structural energy storage devices (SESDs) that not only store electrical energy efficiently but also bear mechanical loads are proposed, possess significant potential to reduce the carbon footprint of the construction industry while advancing towards NZEBs [6,7].

Among various types of SESDs, structural supercapacitors (SSCs) are particularly promising due to their outstanding energy storage capacity, high power density, long cycle life, and most importantly, the robust load-bearing capability. Currently, construction materials possess inherent structural strength and ion transport potential, making them ideal candidates for functionalization as load-bearing and ion-conductive separators in SSCs [8]. Most conventional Portland cement-based SSCs typically employ various alkaline metal salt solutions, such as KOH, KCl, Na_2SO_4 , and LiOH, as electrolyte solutions to enhance ionic conductivity [9–11]. It was also observed that the composite containing Na_2SO_4 achieved a compressive strength of 32.7 MPa and an ionic conductivity of 6.2 mS cm^{-1} when alkaline metal salts were directly mixed with cement slurry [12]. However, the incorporation of alkali metals salts might potentially compromise the durability of cement-based materials [13]. In addition, Lithium trifluoromethane sulfonate (LiOTf), recycled steel slag (SS), and potassium ferricyanide ($\text{K}_3\text{Fe}(\text{CN})_6$) have been used to functionalize Portland cement-based

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Table 1

Chemical composition of cement and fly ash.

Composition	Al ₂ O ₃	CaO	SiO ₂	Fe ₂ O ₃	K ₂ O	MgO	TiO ₂	Na ₂ O	SO ₃	Others.
Cement	8.11	53.59	24.56	3.41	0.70	1.96	0.39	0.19	2.81	4.28
Fly ash	26.73	10.32	41.55	6.92	0.29	1.37	1.66	0.23	0.37	10.56

materials, thereby enhancing ions conductivity and areal capacity for applications in SSCs [14–16]. Hemp fibers were incorporated into magnesium sulfate cement to introduce connected pores, which enhanced electrochemical properties with a notable trade-off in mechanical properties [17]. Meanwhile, current research frequently incorporates organic polymers, mainly including PAA, PAM, and PVA to enhance ionic conductivity of cementitious materials [18–20]. However, polymer aggregation can hinder cement hydration and compromise the compressive strength of cement-based materials. Furthermore, the flammability and toxicity of these polymers pose significant hazard to energy storage devices [21,22]. Metakaolin/montmorillonite-based geopolymers, serving as alternative green binders for construction with superior mechanical property, were also utilized as structural electrolytes in conjunction with various solutions, such as Na₂SO₄ and KOH, and the electrochemical performance is closely related to the ion transport efficiency determined by the internal pore structure [23–25].

Using foaming agents is an alternative method to enhance ionic conductivity through creating a porous structure in cement-based materials [26]. However, this approach often compromises mechanical properties as well. Supplementary cementitious materials, such as fly ash, offer a potential solution to this issue. Although the initial reactivity of fly ash is significantly lower compared to that of cement particles, the hydration kinetics of cement can be enhanced by leveraging heterogeneous nucleation, which catalyzes the nucleation process of hydrates on

fly ash particles by reducing the energy barrier [27]. Moreover, the spherical geometry of fly ash particles can act as ball bearings to enhance the rheological properties of fresh pastes [28]. Fly ash particles with spherical geometries facilitates the homogeneous dispersion and stability of bubbles to refine the pore structure for ions transport, thereby improving the energy storage property of SSCs and stabilizing the electrode-electrolyte interface [29,30]. The hydration process of fly-ash particles, known as pozzolana reaction, can proceed with pore solution and cement hydrates for years after the hardening of pastes [31]. This method can help achieve a balance between electrochemical and mechanical properties in cement-based composites [32–34]. Additionally, the use of fly ash reduces cement usage, further contributing to lower carbon emissions in the cement industry [35].

Besides cement-based composite separators, electrodes also contribute to the overall energy storage performance of structural supercapacitors. In these systems, two-dimensional graphene sheets and modified three-dimensional reduced graphene oxide (rGO) composites (such as rGO/MnO₂, rGO/CuO, rGO/SnO₂, and rGO/Fe₂O₃) are commonly used as electrode materials due to their large theoretical surface area and excellent electrical conductivity [14,16,18,36]. However, the insufficient energy density and intricate manufacturing process of these materials pose challenges for large-scale production and cost reduction in integration with bulk building structures.

Aqueous zinc-ion hybrid supercapacitors have emerged as a

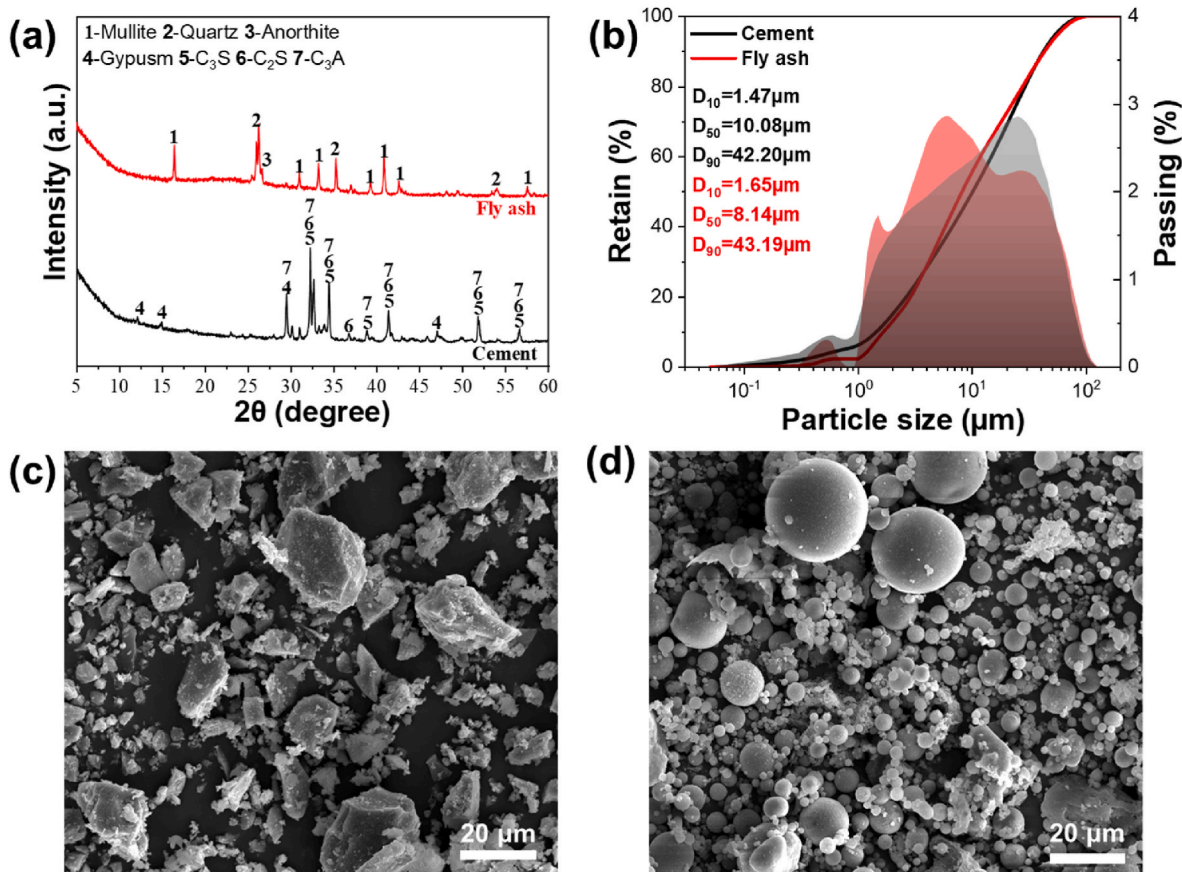


Fig. 1. Characterizations of cement and fly ash. (a) XRD patterns. (b) Particle size distributions. (c) SEM image of cement. (d) SEM image of fly ash.

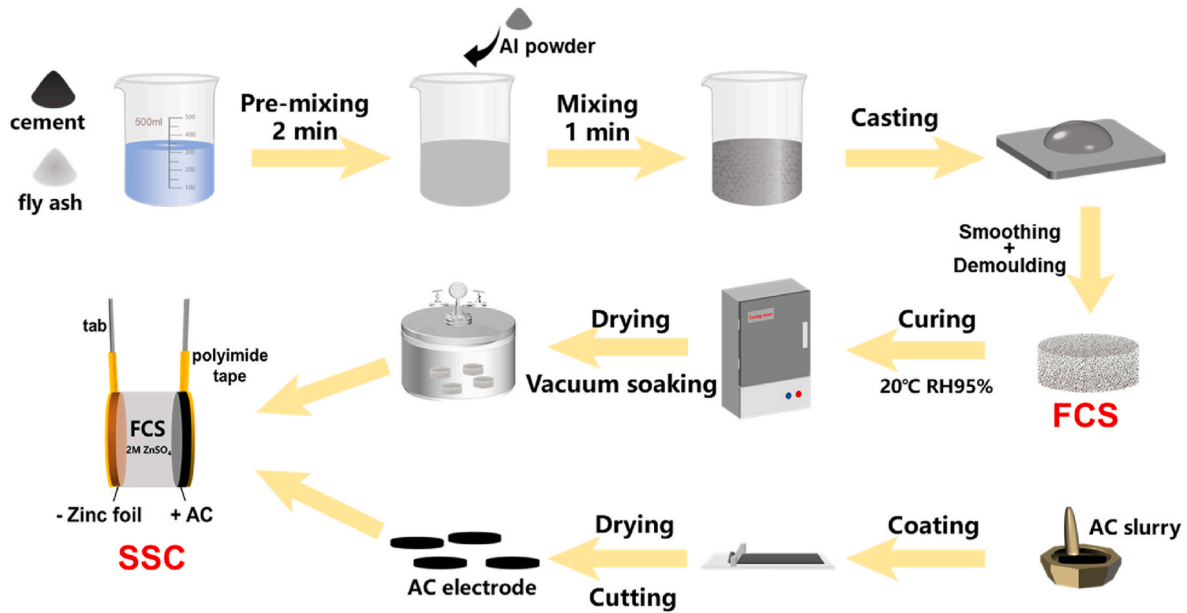


Fig. 2. The preparation of FCS and SSC.

promising energy storage technology, offering inherent safety, abundant raw materials, and exceptional electrochemical energy storage capacity [37]. Zinc metal provides favorable properties such as excellent electrical conductivity (resistivity of $5.91 \mu\Omega \text{ cm}$), high theoretical specific capacity (volumetric capacity of 5855 mAh cm^{-3} and gravimetric capacity of 820 mAh g^{-1}), and a low equilibrium potential (-0.763 V vs. SHE), contributing to high energy density, power density, and long cycle life [38–40]. The integration of SSCs with cement-based composite separators and aqueous electrolytes in structural supercapacitors not only enhances interface compatibility but also holds great potential for advancing large-scale electrochemical energy storage. This novel combination leverages the mechanical robustness and cost-effectiveness of cementitious materials with the superior electrochemical performance of zinc electrodes, paving the way for innovative, scalable, and sustainable energy storage solutions.

In order to address the aforementioned challenges and achieve low-carbon and energy-efficient constructions that align with sustainable development goals, this paper developed a novel class of structural zinc-ion hybrid supercapacitor utilizing porous fly ash-cement composites (FCS). We investigated the microstructure of porous FCS and examined how its porous structure affects energy storage performance, considering the combined effects of curing time and ZnSO_4 soaking. Furthermore, we explore the electrochemical performance of SSCs assembled from FCS, zinc metal anode, and active carbon cathode. Simultaneously, the carbon emissions and energy saving rate of SSCs for structural energy storage were evaluated. This novel combination leverages the mechanical robustness and cost-effectiveness of cementitious materials with the superior electrochemical performance and energy density of zinc electrodes, paving the way for innovative, scalable, and sustainable energy storage solutions. This approach aims to enable large-scale building-integrated energy storage, thereby accelerating the development of net-zero energy consumption buildings.

2. Materials and methodology

2.1. Materials

A Class C fly ash from Nanjing coal power plant and type 42.5R Portland cement were used in this research. The chemical composition of cement and fly ash is provided in Table 1. The mineral components, particle size distribution, and morphology of cement and fly ash

Table 2

Mix proportions and curing conditions of cement-based composites (in mass ratio).

Mixture	Cement	Fly ash	Al powder	Water	Curing age (d)	Soaking condition
CZ1	1	0	0.001	0.5	1	2M ZnSO_4 solution
CZ3					3	
CZ7					7	
CZ28					28	
CH1	0.5	0.5	0.001	0.5	1	Deionized water
CH3					3	
CH7					7	
CH28					28	
FZ1					1	2M ZnSO_4 solution
FZ3					3	
FZ7					7	
FZ28					28	
FH1	0.5	0.5	0.001	0.5	1	Deionized water
FH3					3	
FH7					7	
FH28					28	

particles are shown in Fig. 1. Zinc sulfate heptahydrate ($\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$, AR), acetylene black, N-methyl pyrrolidone (NMP), and polyvinylidene difluoride (PVDF) were purchased from Shanghai Aladdin Bio-Chem Technology Co., Ltd. Activated carbon (AC) were acquired from Jiangsu XFNANO Technology Co., Ltd. The zinc foil with a thickness of 0.05 mm was purchased from Shanghai Weidi Metal New Material Technology Co., Ltd. Al powder was obtained from Shandong Lin Guan Chemical Co., Ltd (90 wt% purity). Furthermore, all the water used in this study was deionized water.

2.2. Preparation of FCS

The water-to-cement ratio of the paste samples adopted in this research was 0.5, and Al powder was used as a foaming agent with a dosage of 0.1 wt% relative to the cementitious material. The prepared specimens were classified into two groups: C (100 wt% cement with 0.1 wt% Al) and F (50 wt% fly ash-50 wt% cement with 0.1 wt% Al), considering the efficient utilization of fly ash without compromising the workability and mechanical properties of the pastes. The cement, fly ash, and water were initially pre-mixed for 2 min, followed by the

addition of Al powder and subsequent stirring for 1 min (see Fig. 2). The pastes were poured into cubic molds (20 mm × 20 mm × 20 mm) for compressive strength tests and cylindrical polyethylene custom molds with a bottom diameter of 12 mm and height of 5 mm for ionic conductivity test and electrochemical properties tests. Other 10 mm × 10 mm × 5 mm blocks were used for the preparation of the large energy storage device. Specimens were demolded after casting for 24 h and then cured in a curing room with (20 ± 2) °C temperature and over 95 % relative humidity for 1, 3, 7, and 28 d. Afterward, specimens were exposed to vacuum drying for 3 d and then soaked in 2M ZnSO₄ solution for 7 d, denoted as CZ (CZ1, CZ3, CZ7, and CZ28) and FZ (FZ1, FZ3, FZ7, and FZ28). Cubic specimens soaked in deionized water for 7 d were used as control groups to compare the compressive strength and microstructure, denoted as CH (CH1, CH3, CH7, and CH28) and FH (FH1, FH3, FH7, and FH28). The detailed mix ratio is shown in Table 2.

2.3. Fabrication of SSCs

To prepare the positive electrode, the slurry was prepared by mixing 80 wt% of active carbon, 10 wt% of acetylene black, and 10 wt% of PVDF in NMP under continuous stirring. The slurry was coated on stainless steel mesh to form a layer with a thickness of 20 μm and dried at 40 °C for 12 h to achieve the AC electrode with a loading of 2.8–3.0 mg cm⁻². The zinc foil in 0.05 mm thick was used as a negative electrode. The electrodes and FCS were fastened by polyimide tape to fabricate SSCs infused with 2M ZnSO₄ electrolyte, as shown in Fig. 2. The nomenclature of the prepared SSCs is in accordance with that of the paste specimens.

2.4. Methodology

2.4.1. Compressive strength

To enhance the efficiency of vacuum soaking while minimizing the solution required, the specimens of different curing ages with the dimensions of 20 mm × 20 mm × 20 mm after soaking in 2M ZnSO₄ solution or deionized water were tested for compressive strength. The compressive strength was determined using a JES-300 concrete compressive strength tester at a loading rate of 0.5 mm min⁻¹. The reported compressive strength is the average value of six samples.

2.4.2. Electrochemical measurements

The electrochemical impedance spectroscopy (EIS) and cyclic voltammetry (CV) tests were performed on the CHI660E workstation at room temperature. The cylinder specimen was sandwiched between two stainless steel foils for ionic conductivity measurement by EIS. The intercept of the curve in the real axis was taken as the bulk resistance R_b (Ω), and the ionic conductivity σ (mS cm⁻¹) of the sample was calculated according to Eq. (1) as follows:

$$\sigma = \frac{L}{S \times R_b} \quad (1)$$

where L and S are the thickness (cm) and cross-sectional area (cm²) of the sample, respectively. The EIS plots of SSCs were recorded over a frequency range from 100 kHz to 0.1 Hz, with the initial voltage set to the open circuit potential. CV experiments were carried out within a working voltage range of 0.2–1.8 V at a scan rate of 10 mV s⁻¹. All galvanostatic charge-discharge and charge-discharge cycle tests for the SSCs were conducted on the LANHE battery test system. The specific capacity Q (mAh g⁻¹), energy density E (Wh kg⁻¹), and power density P (W kg⁻¹) can be calculated based on Eqs. (2)–(4):

$$Q = \frac{It}{m} \quad (2)$$

$$E = \frac{\int IV(t)dt}{m} \quad (3)$$

$$P = \frac{E}{t} \quad (4)$$

Where I , t , m , and $V(t)$ refer to the discharge current (A), discharge time (h), the mass (g) of active materials in the cathode, and discharge voltage (V), respectively.

2.4.3. Microstructure characterizations

The phase assemblage of the cementitious materials was characterized by X-ray Diffraction (XRD). A Bruker D8 Advance X-ray Powder Diffractometer, equipped with CuKα radiation operating at 40 kV and 30 mA, was used. Data were collected over a 2θ range of 5°–60° using continuous scanning.

Thermogravimetry analysis (TGA) was carried out on a NETZSCH STA 449F5 instrument. The measurement was performed from 30 °C to 1000 °C at a heating rate of 10 °C min⁻¹ under a N₂ atmosphere.

Scanning electron microscopy (SEM) using a FEI 3D microscope was employed to observe the microstructure of FCS samples. Additionally, energy dispersive X-ray spectroscopy (EDX) was conducted to analyze the elemental composition of the samples. Multiple points within the microstructure were selected for EDX analysis to ensure a comprehensive understanding of the elemental distribution and to identify the presence of hydration products and the reaction products between the cement and electrolyte across different regions of the FCS samples.

Mercury Intrusion Porosimeter (MIP) and X-ray Computed Tomography (X-CT) were employed to determine the pore structure of the specimens. For the MIP test, dried pieces of 1–3 mm in length were analyzed using an Auto Pore IV 9500 porosimeter, with the intrusion pressure progressively increased from 0 to 200 MPa. Additionally, dried cylindrical samples were scanned using a ZEISS Versa XRM500 X-CT system. The working voltage was 190 kV and the output current was 0.15 mA. Interactive thresholding using AVIZO software is utilized for segmenting the skeleton and pores, as shown in Fig. S1. The equivalent pore network model represents an organic combination of spatial dimension and two-dimensional image, effectively characterizing microscopic pore and throat sizes and the topological properties [41].

2.4.4. Calculation of CO₂ emission and energy saving rate

The embodied carbon EC (kg CO₂e t⁻¹, representing the equivalent CO₂ emissions per ton of material) serves as a comprehensive indicator of greenhouse gas emissions [42], encompassing CO₂ emissions along with other greenhouse gases, and can be quantified using the carbon coefficient [43]. The specific calculation formula is shown in Eq. (5).

$$EC = \sum_{i=1}^n (Q_i \times CC_i) \quad (5)$$

Where:

Q_i is the quality of raw material i , kg.

CC_i is the carbon coefficient of raw material, kg CO₂e t⁻¹, and the CC_i of electricity and each component material is presented in Table S1.

The social benefit of SSCs was evaluated by calculating the energy saving rate (SR) after using SSCs for residential electricity supply, as shown in Eqs. (6)–(8).

$$SR = \frac{EngS}{EngC} \times 100\% \quad (6)$$

$$EngS = EngD \times ConC \times R \quad (7)$$

$$SR = \frac{EngD \times ConC \times R}{EngC} \times 100\% \quad (8)$$

Where:

$EngC$ represents the residential per capita electricity consumption in kWh/capita, obtained from data by the National Bureau of Statistics of China, the United States Energy Agency, and the European Commission

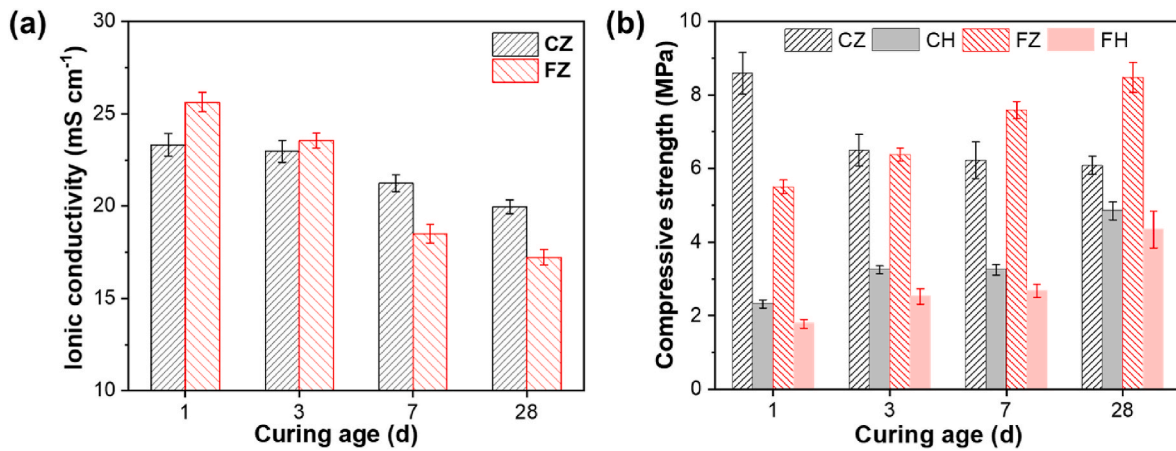


Fig. 3. (a) Ionic conductivity and (b) compressive strength of FCS at different curing ages.

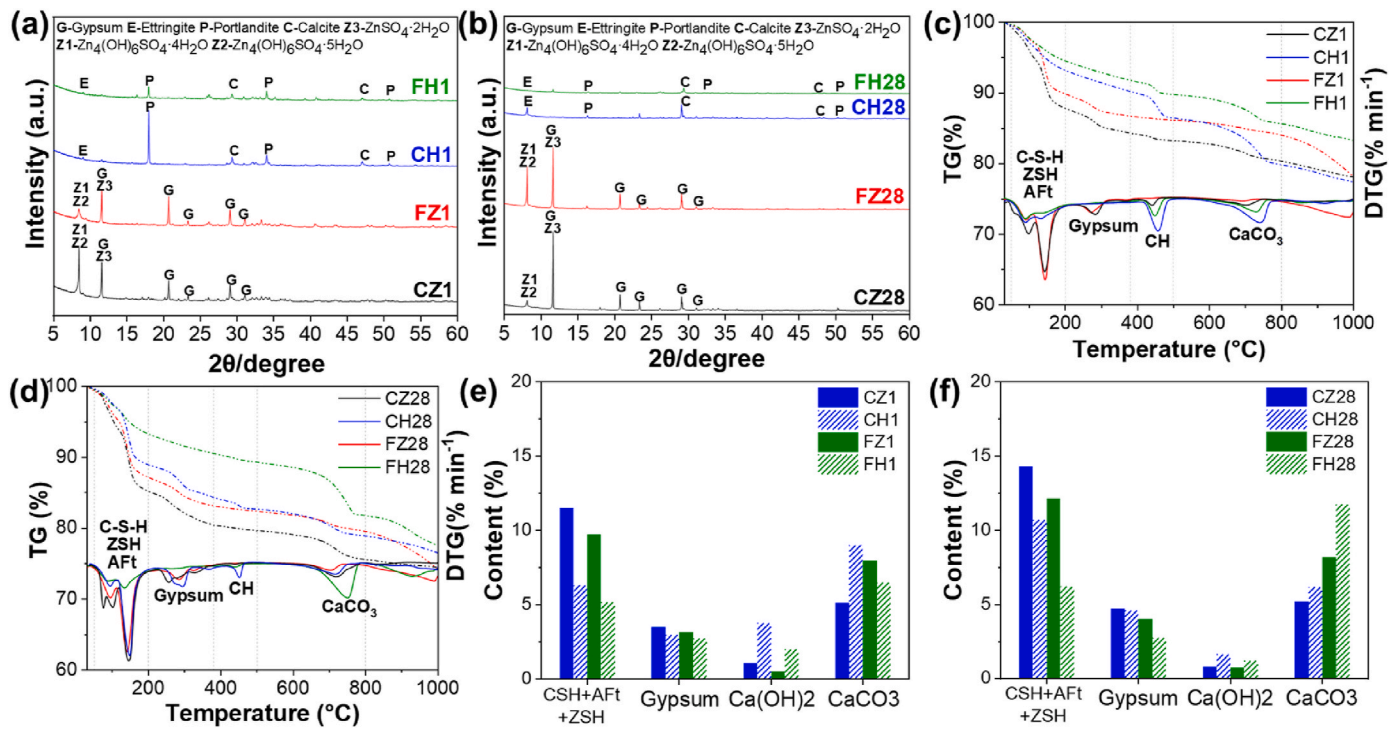


Fig. 4. Mineralogy and chemical compositions of FCS. (a–b) XRD patterns. (c–d) TG/DTG profiles. (e–f) Contents of products.

[44–46].

EngS represents the per capita electricity storage, kWh/capita.

EngD represents the energy density of SSCs, kWh m⁻³.

ConC represents the per capita concrete consumption of residential buildings, m³/capita, obtained from data provided by the National Bureau of Statistics of China and World Bank Open Data [44,47].

R is the proportion of SSCs replacing conventional concrete, %.

3. Results

3.1. Ionic conductivity and compressive strength of FCS

The ionic conductivity for FCS with 2M ZnSO₄ of different curing ages is shown in Fig. 3(a). Generally, the ionic conductivity decreases with increasing curing age. Adding fly ash significantly increases ionic conductivity at early curing age (1 d and 3 d), with FZ1 achieving a peak value of 25.6 mS cm⁻¹. However, this enhancement is diminished over

time as the microstructure densifies that impedes the ion migration.

Fig. 3(b) presents the effects of ZnSO₄ soaking on the compressive strength of FCS at various curing ages. Samples soaked in ZnSO₄ solution exhibit significantly higher compressive strength compared to those soaked in deionized water, stabilizing with increasing curing age. ZnSO₄ reacts with hydration products to improve compressive strength, reaching values approximately 210 %–500 % higher than those of conventional Al powder-foamed concrete [48]. Notably, ZnSO₄ soaking continually improves the compressive strength in FCS containing fly ash, whereas the compressive strength of FCS without fly ash slightly decreases with curing age.

3.2. Microstructural of FCS

3.2.1. Mineral and chemical compositions

The mineral composition of FCS of different curing ages determined by XRD is shown in Fig. 4(a–b) and Fig. S2. The predominant phases of

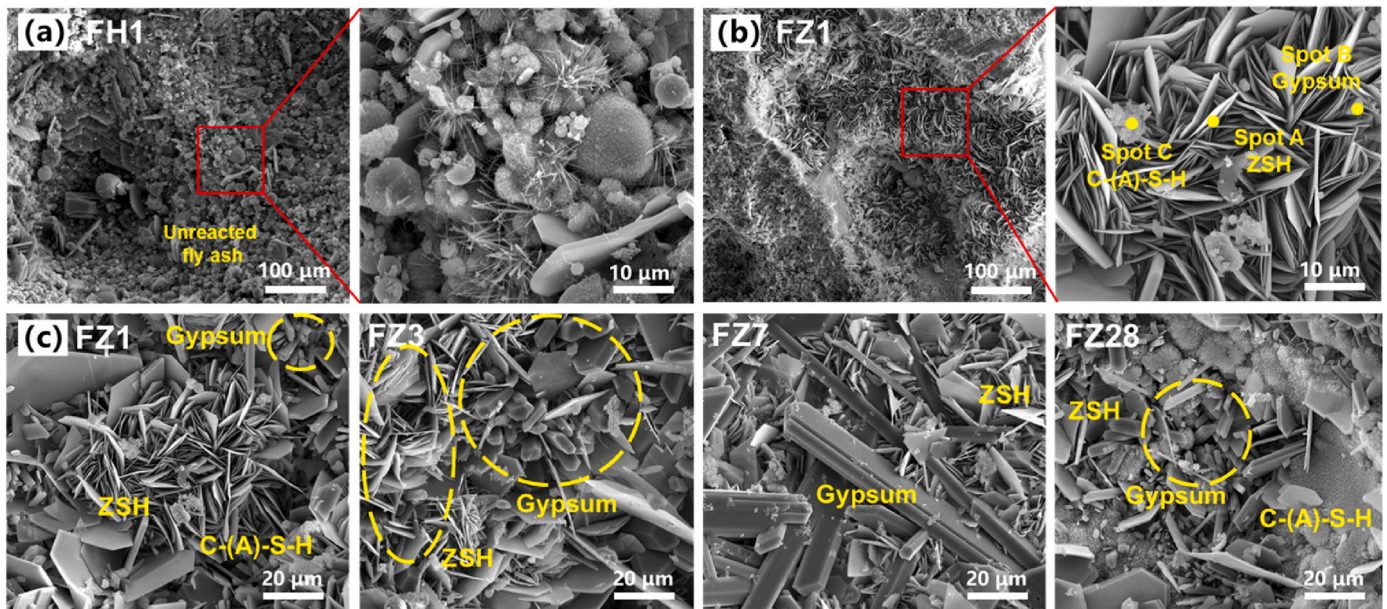


Fig. 5. SEM images of FCS. (a) FH1. (b) FZ1. (c) FZ group of various curing ages.

the sample soaked in deionized water are ettringite (Aft), portlandite, and calcite. These arise from cement hydration, the pozzolanic reaction of fly ash, and the carbonation of cement hydration products exposed to air. The peak intensity of portlandite decreased with increased curing age, indicating its consumption due to the pozzolanic reaction with the presence of fly ash [49]. In FCS samples soaked in ZnSO_4 solution, gypsum and zinc sulfate hydroxide hydrate ($\text{Zn}_4(\text{OH})_6\text{SO}_4 \cdot x\text{H}_2\text{O}$, designated as ZSH) are detected. This suggests that these compounds are

formed from the reaction of ZnSO_4 and $\text{Ca}(\text{OH})_2$ [50], with the latter disappearing in these systems. $\text{ZnSO}_4 \cdot 2\text{H}_2\text{O}$ is also observed, likely precipitating during the drying process.

The TG/DTG curves of FCS cured for 1 d and 28 d are shown in Fig. 4 (c–f). The content of C-S-H, Aft, and ZSH in cement paste soaked in ZnSO_4 solution is significantly higher than in samples soaked in deionized water, supporting the observed higher compressive strength (Section 3.1). As shown in Fig. 4(e–f), the increment of CSH, ZSH, and

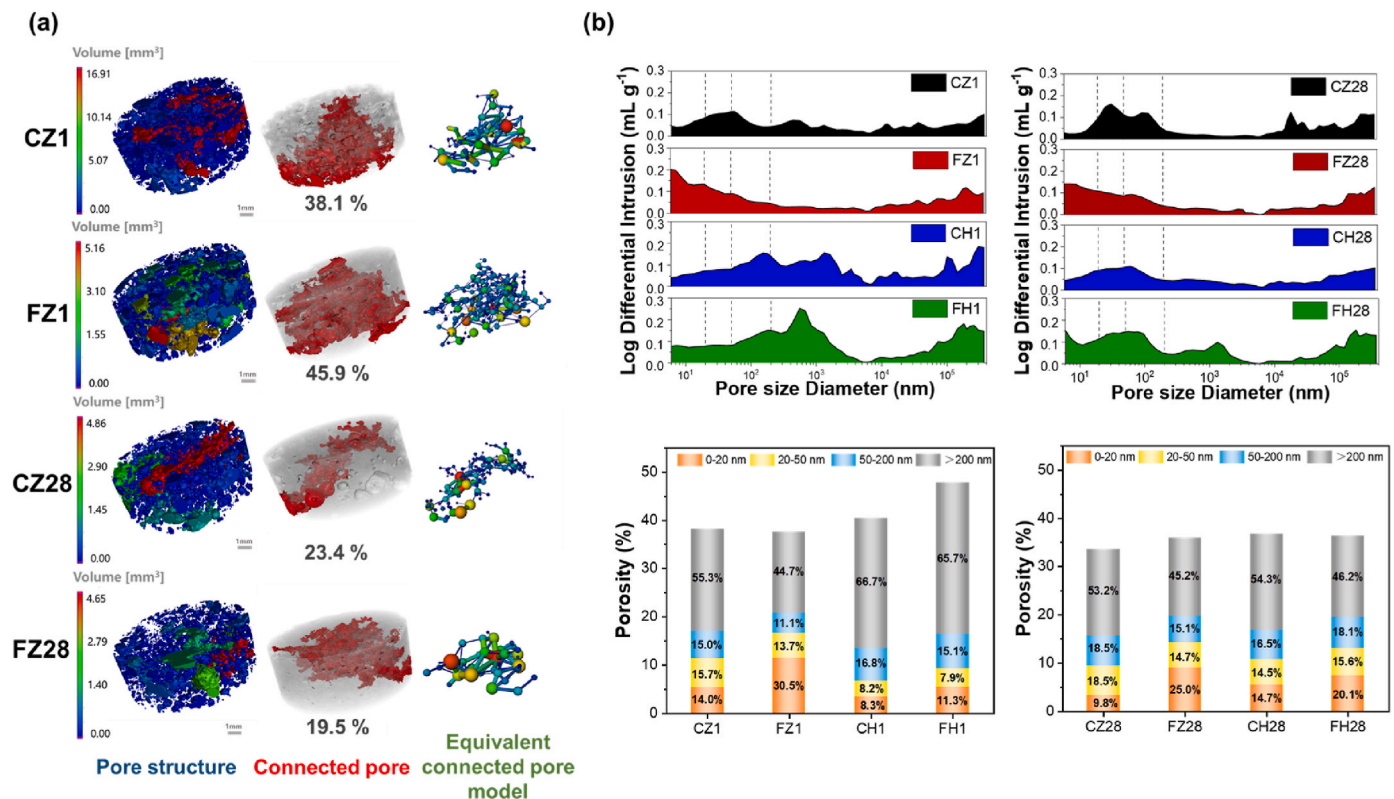


Fig. 6. Pore structure characterization of FCS. (a) 3D pore visualization and connected porosity via X-CT. (b) Pore size distribution and relative volume fraction via MIP.

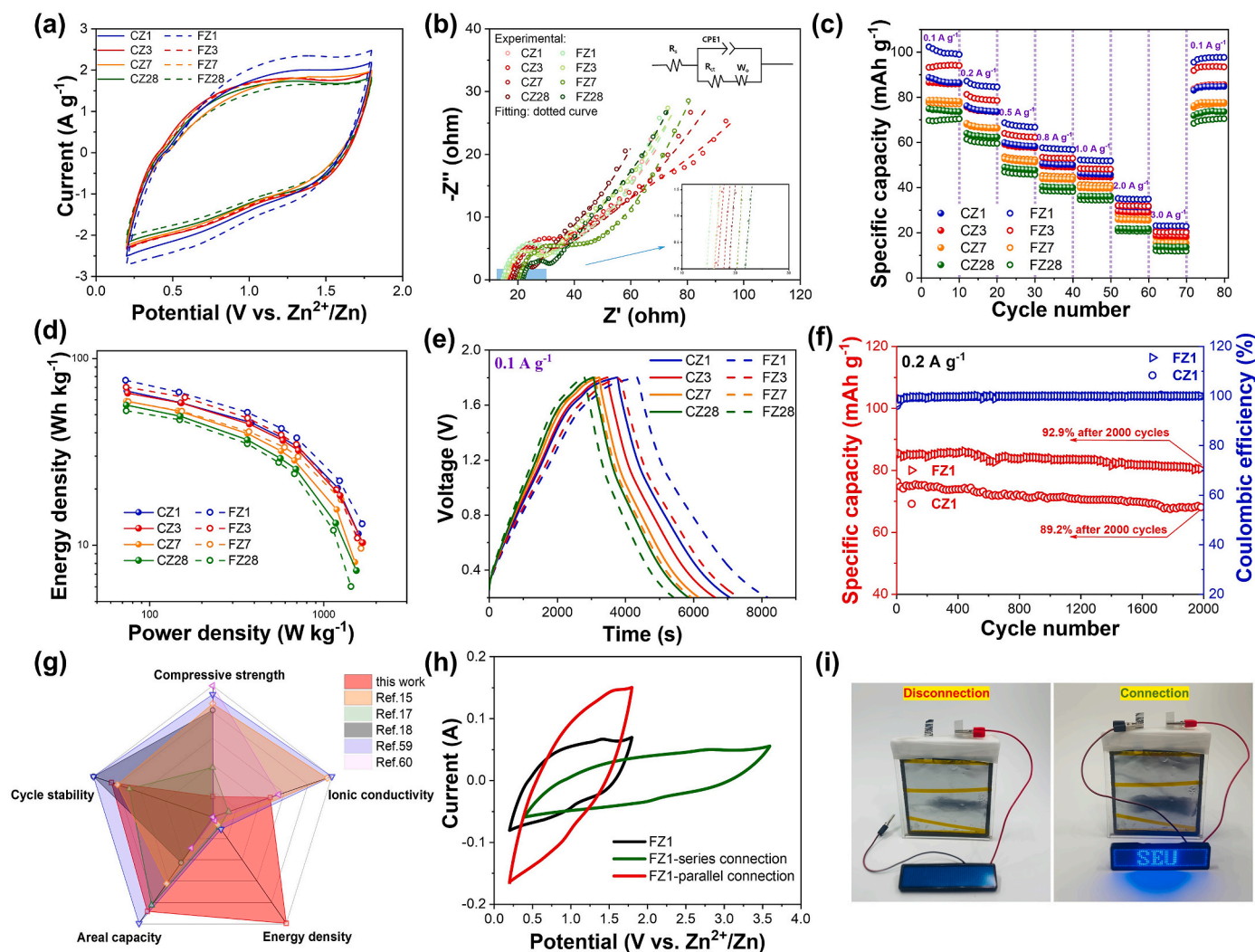


Fig. 7. Electrochemical performance of SSCs with varied curing ages. (a) CV curves at a scan rate of 10 mV s⁻¹. (b) EIS spectra. (c) Rate performance. (d) Energy density vs. power density. (e) Galvanostatic discharging/charging curves. (f) Cyclic stability at 0.2 A g⁻¹. (g) Radar plots for performance comparison of this work and related references [15,17,18,59,60]. (h) CV curves of larger SSCs (1 and 2100 cm²) at 1 mV s⁻¹. (i) Photos of blue LED lighted by one 100 cm² cell. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

gypsum content in CZ group decreases with increased age, explaining the diminishing improvement in compressive strength of ZnSO₄ soaked samples with prolonged curing age. As the hydration reaction progresses, the content of Ca(OH)₂ in the cement system decreases, leading to a reduction in its reaction with ZnSO₄ [51]. Fly ash, although possessing lower hydration activity initially, will gradually hydrate over time. In other words, the pozzolanic effect of fly ash can facilitate the secondary hydration reaction, thereby interacting with ZnSO₄ to maintain the level of strength enhancement.

3.2.2. Morphology

Fig. 5 depicts the morphology characteristics of fly ash-cement paste soaked in deionized water (Fig. 5(a)) and ZnSO₄ solution (Fig. 5(b)). A rough surface and a significant amount of unreacted fly ash are observed in FH1 (Fig. 5(a)). Conversely, it can be observed that FZ1 (Fig. 5(b)) exhibits a uniform and dense morphology, with a significant abundance of finely distributed hexagonal products containing zinc (such as ZSH) and gypsum on the matrix surface (energy dispersive spectrometer analysis in Fig. S3), which effectively enhanced the strength. With the progression of curing age, there is a gradual increase in gypsum and C-(A)-S-H gels (see Fig. 5(c)), leading to subsequent enhancement in compressive strength. However, the presence of dense products

hindered ion migration [20].

3.2.3. Pore structure

The pore structure characteristics are closely related to the ion transport and mechanical strength of FCS. The overall pore structure of FCS was characterized by the X-CT (pore diameter >50 μm) and MIP (pore diameter ranging from 5 nm to 300 μm) techniques, as shown in Fig. 6. The porosity of the samples soaked in ZnSO₄ solution is significantly reduced (see Fig. 6(a) and Fig. S4), which also supports to the enhanced compressive strength (Section 3.1). As curing age increases, porosity reduces, leading to a corresponding decrease in ion transport efficiency [52]. Moreover, the addition of fly ash contributes to forming a microstructure with more uniformly distributed pore sizes (see Fig. S4). This can be attributed to the unique spherical geometry of fly ash, acting as ball bearings during the foaming process, facilitating the breakup of large bubbles and ensuring the uniform dispersion of air voids [53]. As shown in Fig. 6(a), the proportion of connected pores in the cement-based composite is observed to be higher at an early age and subsequently decreases with increasing curing age. In comparison to CZ1, although the porosity of FZ1 is slightly lower, it exhibits a higher proportion of connected pores and a more uniform network distribution. This characteristic can effectively mitigate compressive strength loss

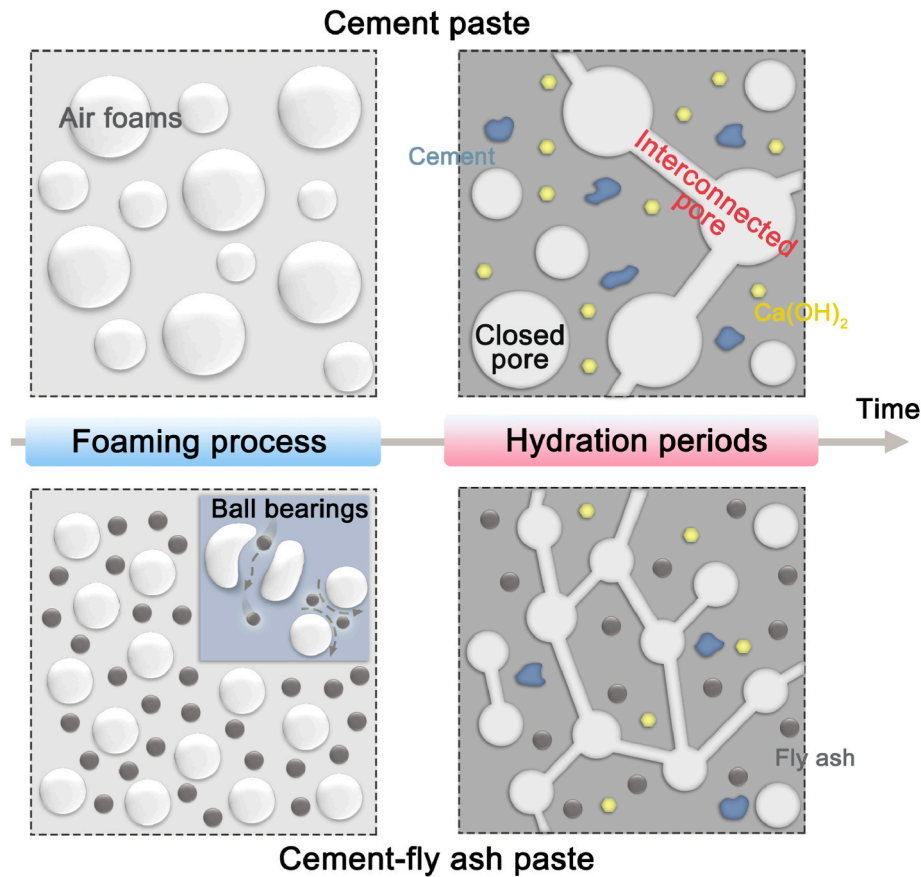


Fig. 8. Schematic illustrations of fly ash influence mechanisms in FCS.

and enhance ionic conductivity. Generally, the microscopic pore structure of cement-based materials is important for ion migration [16,52]. Fig. 6(b) presents the pore size distribution and porosity via MIP. The pores exceeding 200 nm in the samples with ZnSO₄ solution decreased significantly, which can be attributed to the pore refining effect of ZSH and gypsum formation. In particular, pores with size smaller than 20 nm increases significantly after adding fly ash, which is essential for enhancing both the ionic conductivity [16] and compressive strength [54].

3.3. Electrochemical performance of SSCs

3.3.1. CV and EIS analysis

The electrochemical performance of SSCs comprising FCS and 2M ZnSO₄ electrolyte, with zinc anode and AC cathode assembled on both sides, was comprehensively characterized. As illustrated in Fig. 7(a), the CV curve exhibits a nearly rectangular shape, indicating the performance of electrochemical double layer capacitance [55]. The area of the CV curves decreases with increasing curing age, with FZ1 exhibiting the maximum CV curve area, suggesting that partial replacement of cement with fly ash enhances capacitive behavior. Fig. 7(b) displays the EIS spectra of the SSCs with FCS at varied curing ages. The bulk resistance (R_b) of SSCs was determined from the EIS analysis. With the addition of fly ash, the R_b of FZ1 is reduced to 14.7 Ω . Generally, R_b increases with curing age.

3.3.2. Rate performance

The rate performance of SSCs at different current densities from 0.1 to 3.0 A g⁻¹ is shown in Fig. 7(c). The discharge capacity gradually decreases with increasing curing age. The FZ group exhibits higher discharge capacities than the CZ group during the initial 1–3 d of curing,

whereas the CZ group demonstrates higher discharge capacities from 7 to 28 d of curing, consistent with ionic conductivity trends. Notably, FZ1 displays discharge capacities of 102.4, 87.2, 68.7, 57.6, 52.2, 35.1, and 23 mAh g⁻¹ at different current densities. When the current density returns to 0.1 A g⁻¹, FZ1 still retains a discharging capacity of 97.7 mAh g⁻¹. These results indicate that FZ1 possesses the best rate performance among all the composite separators tested, attributed to its highest ion conductivity.

For further verification, Fig. 7(d) presents the relationship between energy density and power density for various FCS samples with different compositions and curing ages. Overall, the figure shows that the energy density decreases as the power density increases for all samples. Additionally, energy density tends to decrease with increasing curing age for both CZ and FZ samples. Notably, FZ1 shows the highest specific energy density among all the samples, reaching 73 Wh kg⁻¹ at a power density of 76.3 W kg⁻¹ and remaining at 10.9 Wh kg⁻¹ when the power density increases to 1.3 kW kg⁻¹.

Fig. 7(e) demonstrates the galvanostatic charge/discharge curves of the SSCs under the current density of 0.1 A g⁻¹. The approximately symmetric triangular shape of the curves indicates the reversible electrochemical characteristics of the devices [18].

3.3.3. Long-term cycling performance

The long-term cycling performance of SSCs at a current density of 0.2 A g⁻¹ is shown in Fig. 7(f). FZ1 demonstrates exceptional cycling stability with a remarkable capacitance retention rate of 92.9 % and coulombic efficiency of nearly 100 % after 2000 charge/discharge cycles. In contrast, the CZ1 group exhibited a lower capacitance retention rate of 89.2 %. This difference may be attributed to the heterogeneous pore structure of CZ1, leading to uneven electrolyte wetting and non-uniform Zn plating [56]. In addition, zinc ions may not be easily

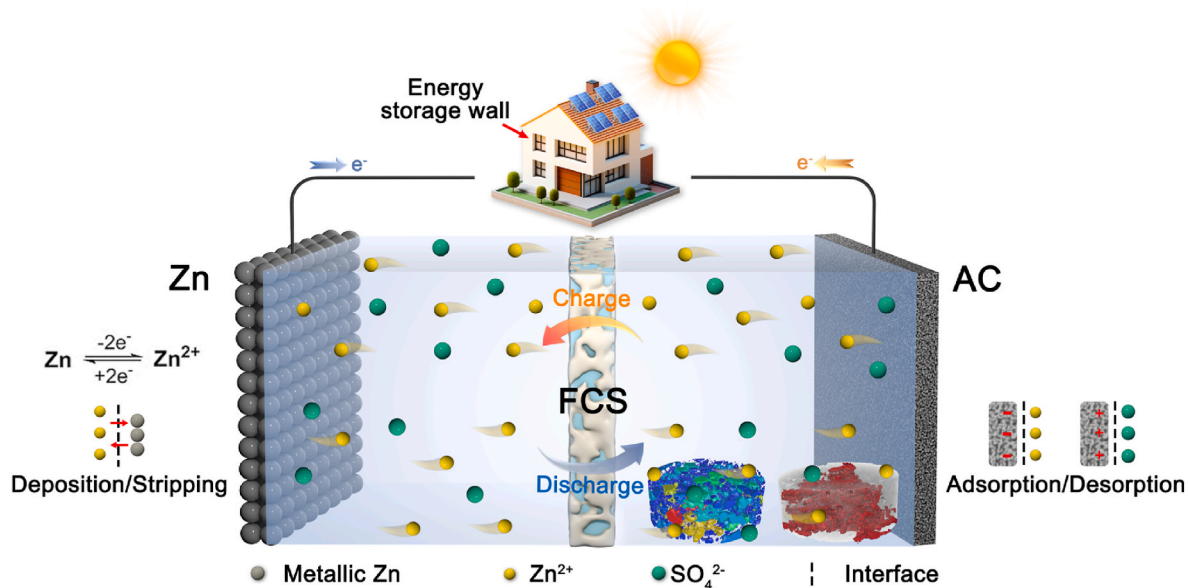


Fig. 9. Schematic diagrams of the energy storage mechanism in fly ash-cement based SSCs.

deposited where the cement-based composites separator contacts the cathode and instead primarily deposit in the large pore cavities of CZ1. This selective deposition can induce the vertical growth of zinc ions, forming dendrites over time [57]. Fig. S5 illustrates the polarization voltage curves of Zn plating/stripping of symmetrical batteries using zinc electrodes at a current density of 1.0 mA cm^{-2} with the capacity of 1 mAh cm^{-2} . The result shows the lower polarization of FZ1 than CZ1, indicating the better compatibility of the zinc anode.

The addition of fly ash in FZ1 create a uniform pore structure with optimal pore size, which reduces internal resistance and polarization voltage, preventing local uneven current density and thereby enhancing the long-term cyclic stability [58]. FZ1 samples soaked in ZnSO_4 solution for various durations (up to 120 d) were assembled into SSCs to evaluate long-term stability and reliability for the structure energy storage (Fig. S6). The results demonstrate a sustained enhancement in compressive strength, ensuring the safety of construction energy storage within the building framework while maintaining acceptable electrochemical performance.

Fig. 7(g) compares the performance of this work with related research, highlighting its exceptional energy density, high areal capacity, and outstanding cycle stability. These attributes suggest its potential as a structural material in conjunction with renewable energy sources.

3.3.4. Demonstration of structural energy storage using larger SSCs

The development of bulk structural supercapacitors involves enlarging electrode area to increase the active substances loading, thereby enhancing the device's energy storage capacity. This approach is particularly well-suited for integration with structural elements such as walls, beams and pillars. As illustrated in Fig. 7(h), two 100 cm^2 cells using FZ1 can achieve a voltage of 3.6 V when connected in series and exhibit elevated capacity when connected in parallel. A single bulk FZ1 device charged at 1.8 V successfully powered a blue light board steadily, as shown in Fig. 7(i). These results demonstrate FZ1's significant potential as a high energy density power source for future structural energy storage devices, showcasing its capacity to integrate with solar energy and reduce the carbon footprint of buildings.

4. Discussion

4.1. Effects of fly ash on the ion transport within FCS

Fly ash influences the ionic transport within FCS through multiple mechanisms, including alterations to pore structure and cement hydration products. As illustrated in Fig. 8, firstly, during the foam formation process, fly ash particles with spherical geometries act as ball bearings, facilitating the homogeneous dispersion of bubbles and refining the pore structure [31]. In addition, due to the much lower activity of fly ash compared to cement clinkers, at the early stages of hydration, fly ash primarily behaves like inert micro-aggregates [61,62]. During hydration, FZ1 generates sparsely distributed hydration products diluted by these micro-aggregates, resulting in increased connected pore and micro-porosity (Fig. 6) that benefits ion conductivity (see Fig. S7). Gypsum and ZSH grow within the pore structure, further refining it and strengthening the overall matrix, as evidenced by the nearly twofold improvement in compressive strength (Fig. 3).

Regarding the cement hydration products, the replacement of cement clinkers with fly ash generates less Ca(OH)_2 and reduces the alkalinity of the pore solution during the early hydration [63]. Consequently, XRD and TG analysis (Fig. 4) show that less ZSH and gypsum are formed, leading to a higher concentration of free zinc ions in the pore solution of FZ1, which becomes another contributor to enhance the ionic conductivity, achieving a peak value of 25.6 mS cm^{-1} . However, as curing progresses, the slowly on-going pozzolanic reaction between fly ash and cement hydration products results in the formation of more hydration products [49]. The accumulation of ZSH and gypsum would gradually narrow the pores sizes and obstruct ion transport. This explains the reduction in rate and long-term cycling performances of SSCs at late stages.

4.2. Energy storage mechanism

The energy storage mechanism of SSCs is demonstrated in Fig. 9. The zinc anode proceeds with the reversible deposition/stripping reaction of Zn/Zn^{2+} during charging/discharging process. This Faradaic process provides SSCs to achieve high energy density. The AC cathode primarily functions via an electrochemical double layer capacitance mechanism, where charge storage occurs through the physical adsorption/desorption of Zn^{2+} and SO_4^{2-} ions at the electrode-electrolyte interface. This

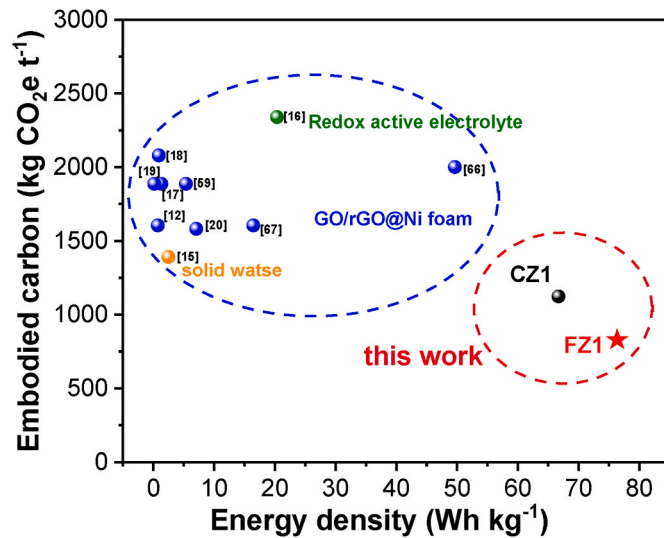


Fig. 10. Embodied carbon comparison of SSCs developed in this work and other cement-based supercapacitors [12,15–20,59,66,67].

process contributes to high power density and cycle stability [39]. The charging and discharging ability of SSCs is primarily determined by the highly connected and uniformly distributed pore structure of FCS, which

consists of the macropores produced by Al powder and the micropores formed through the hydration and reaction products after the refinement of zinc sulfate electrolyte. This refined pore structure enhances the efficient ion transport while improving the electrolyte-electrode interface, thereby extending the service life of the electrodes and ultimately prolonging the cycle life of SSCs. Moreover, the high amount of ferrite phase and the presence of unburned carbon phase in fly ash may expand the electrochemical double layer capacitance behavior at the interface between the electrolyte and AC electrode [64]. This phenomenon warrants further investigation.

4.3. Sustainability analysis on the SSCs

4.3.1. Reduction in CO₂ emission

The energy density of conventional cement-based structural energy storage devices is relatively low. The addition of special water-in-salt electrolytes can significantly expand operation voltage and increase energy density of structural supercapacitors [14,65]. However, the high cost poses a barrier to their large-scale application in construction energy storage. Given the outstanding energy density and cyclic stability of the novel SSCs developed based on FCS, a preliminary assessment of the device's CO₂ emission has been conducted, considering both the raw materials and the manufacturing process. The results, shown in Fig. 10, illustrate that the carbon emissions of cement-based structural energy storage are significantly reduced by SSCs. This reduction is attributed to the combination of the abundant availability of zinc and AC electrodes, a

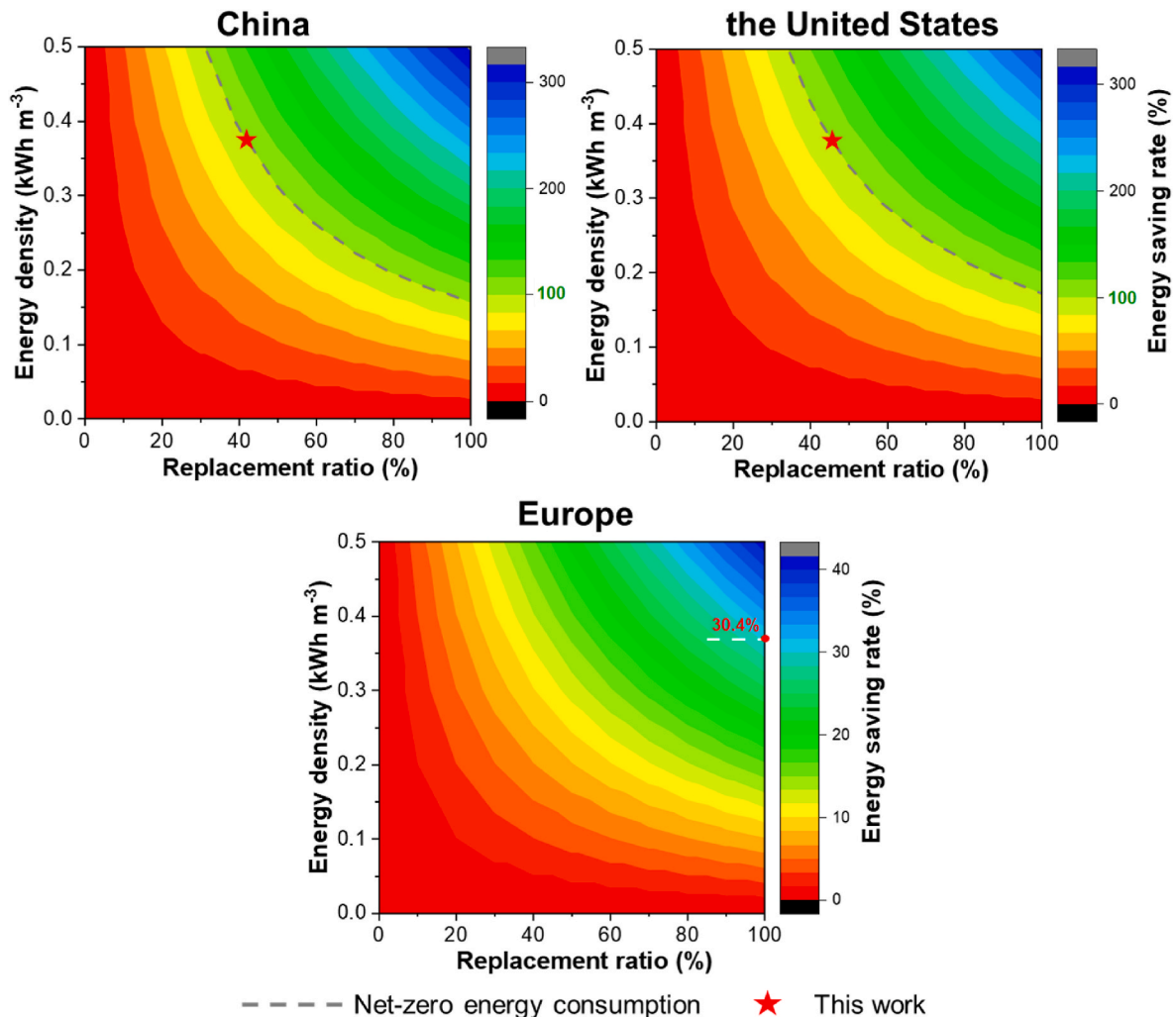


Fig. 11. Estimation of energy saving rate for China, the United States, and Europe when SSCs are integrated into buildings replacing existing concrete components.

cost-effective approach for pore-forming, as well as the incorporation of substantial quantities of solid wastes such as fly ash. In addition, the prepared device serves as an energy storage component that releases electricity to compensate for residential electricity consumption, further facilitating carbon reduction. When integrated into structural elements and combined with renewable energy, SSCs possess the potential to achieve full life cycle negative carbon emissions for buildings.

4.3.2. Net-zero energy consumption buildings

One important application of the developed SSCs device is its use as structural components for energy storage, thereby enhancing the efficiency of renewable resource utilization and contributing to the realization of net-zero energy consumption buildings. Fig. 11 shows an estimated energy saving rate for China, the United States, and Europe when SSCs are integrated into buildings replacing existing concrete components. This estimation assumes an energy density range for SSCs from 0 to 0.5 kWh m⁻³ and a replacement rate from 0 to 100 %. The net-zero energy consumption curves are indicated in dash lines, demonstrating that a 100 % energy saving rate can be achieved at appropriate replacement ratios and energy densities. With the energy density achieved in this study (i.e., 0.37 kWh m⁻³), the required replacement ratio of SSCs to completely offset the electricity consumption is 42.3 % in China and 46.6 % in the United States. However, in Europe, where per capita electricity consumption is 7.5 times higher than in China and 4.2 times higher than in the United States, and per capita concrete consumption is relatively low, the energy saving rate is limited to less than 30.4 % at the current energy density. Nonetheless, SSCs serving as a structural component for energy storage can significantly contribute to the goal of net-zero energy consumption globally. This impact will become more substantial with an increase in the energy density of SSCs.

5. Conclusion

In this study, we developed a novel zinc-ion hybrid structural supercapacitor based on porous fly ash-cement composites, achieving a superior specific capacitance of 102.4 mAh g⁻¹ at 0.1 A g⁻¹, high energy density of 73 Wh kg⁻¹, and remarkable cycling stability with a capacitance retention rate of 92.9 % after 2000 cycles. The porous fly ash-cement composites containing 50 wt% fly ash provide sufficient compressive strength and exhibit high ionic conductivity, preventing internal short circuit effectively as separator and load-bearing structure. The incorporation of ZnSO₄ electrolyte enhances both the ionic conductivity and compressive strength of the composites. The formation of gypsum and ZSH fills the macro pores and refines the pore structure, thereby constituting the primary mechanism for enhancing strength.

The addition of fly ash significantly improves the energy storage performance of SSCs by creating a uniformly distributed and refined pore structure during the early curing stage, attributed to the ball-bearing effect and pozzolanic reactions. Efficient utilization of fly ash reduces not only significantly reduces CO₂ emission and lowers the cost of this energy storage device, but also offers a breakthrough approach for the high-value use of substantial solid wastes. Therefore, the strategy of employing cement and solid wastes for structural energy storage holds a great potential for large-scale energy storage and carbon footprint reduction, thereby facilitating the achievement of net-zero energy consumption in buildings worldwide.

CRedit authorship contribution statement

Ruidan Liu: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Pan Feng:** Writing – review & editing, Project administration, Methodology, Funding acquisition. **Zhaolong Liu:** Methodology, Investigation. **Long Yuan:** Investigation. **Guanghui Tao:** Investigation. **Zhenqi Yu:** Methodology. **Xiangyu Meng:** Investigation. **Jian Chen:** Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cemconcomp.2025.105960>.

Data availability

Data will be made available on request.

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